

Nick Mak

Full Stack Software Engineer | New York, NY
(415) 609-9900 | Portfolio: [nickmak.com](#) | [nicknickmak@gmail.com](#) | LinkedIn: [nick-l-mak](#)

Summary

Software Engineer with a strong focus on scalable distributed systems and performance optimization. Skilled at breaking down ambiguous challenges, driving clarity across teams, and delivering high-impact technical solutions from concept to production.

Work Experience

Software Engineer — Microsoft | April 2022 - July 2025 | New York, NY

- Developed and maintained a large-scale, full-stack web service infrastructure, ensuring high availability, performance, which **reduced latency by 33%**.
- Created full-stack features using **React**, **Angular**, **.NET C#**, **MS SQL Server**, **Azure Cosmos DB**, **Redis**, and **Azure Microservice Architecture**.
- Led the UI development for an internal Ads flighting experimentation product, enhancing UI/UX functionality and optimizing the UI experience, by implementing virtual scrolling and browser caching.
- Created **RESTful APIs** to remove manual **SQL Server** database queries, improving maintainability and minimizing faults and risks for frequent writes of new data requirements.
- Completed UI codebase migration from AngularJS to Angular, improving maintainability and minimizing bugs.
- Executed mass refactoring of legacy **.Net C#** Backend codebase for updated compatibility, documentation, and reusability by removing redundancies and expired business logic.
- Utilized **Azure Copilot** to develop an **AI RAG Assistant** for the web service, improving productivity and users' workflow.
- Built the UI for a 1-click checkout feature on the MSN Shopping page using **React**, from start to finish. Feature takes user provided and saved wallet information to autofill and streamline the user's checkout process for any purchasing experience.
- Integrated **end-to-end** and **unit testing** with **Selenium**, **Moq** and **Jest** for **90%** code coverage in both Frontend and Backend codebases to ensure code quality and reliability.
- Collaborated with directors, data scientists, designers and product managers to implement new features and improve existing ones, ensuring alignment with user needs and business goals.

Software Engineer Intern — Microsoft | June 2021 - September 2021 | Remote

- Built a batch-processing data pipeline for centralizing onboarding Ads Campaign data for BingAds using Azure Data Lake.
- Generated high-visibility Ads Globalization and OKR metrics with **Databricks** in **Scala** to demonstrate the new pipeline.
- Utilized **Apache Spark** for **data mining**, Azure Data Factory for scheduling, and **Power BI** for data visualization.

Education

University of California - San Diego | Bachelor's of Science, Computer Science | September 2018 - December 2021

Certification

Google Cloud Platform | Associate Cloud Engineer | In Progress | November 2025

Technical Projects

PowerAI | OpenAI, RAG, Prompt Engineering, Python, FastAPI, React, FAISS, PostgreSQL

- Built a GenAI full-stack service that utilizes **OpenAI** to simplify users' weightlifting and exercise data onboarding and provide useful metrics for progress tracking and motivation.
- Utilizes a **hybrid RAG retrieval system** using **FAISS** to provide tangible data for potentially fuzzy/obscure user entries, reducing the complexity for the entry format and simplifying user experience.
- Supports a personalized **AI assistant** trainer that uses user's exercise data for tailored feedback.

Peer Pressure | React, Next.js, Typescript, Prisma, PostgreSQL, NextAuth, TailwindCSS, WebSockets

- Built a full-stack social accountability app from 0→1 using **Next.js** and **TypeScript**, enabling users to create daily task lists and view real-time progress from friends.
- Implemented real-time task updates and completion timestamps using a custom **WebSocket** server.
- Designed secure authentication and user-profile system with **NextAuth (JWT)**, supporting persistent sessions and personalized task management.

Technical Skills

- Experienced in developing scalable software solutions using various programming languages and technologies.

Programming Languages

Typescript Javascript C++ C# Java Python HTML5 SASS

Frameworks/Technologies

React Next.js Angular Node.js Django .NET WebSockets OpenAI Azure Microservices GCP Azure Cosmos DB FAISS
RAG Prompt-Engineering PostgreSQL